



Responsible AI Planning and Decision Making

Introduction



Example: Parcel Delivery



Example: Asthma Management

Source: AIHA 2006

Source: American International Health Alliance. BRONCHIAL ASTHMA - Clinical Practice Guideline for General Practitioners. 2006.

<http://www.aiha.com/en/ResourceLibrary/Products/Curricula/pdf/CPG%20Asthma%20EN.pdf> [Accessed 6 Aug 2010]



Example: Characters in Games

Source: <https://aiwarriorswebsite.github.io/AIWarriors/>

Source: www.battle.net/wow

Source: stardewvalley.net

Source: dota2.com

Example: Assistive Robots

Source: Romibo

Source: www.cyberbotics.com

Source: Aldebaran



Example: Scientific and Industry Robots

Source: IMDA



Example: Warehouse Management

- How does Amazon do it?

<https://youtu.be/a77XyUI-zXo>

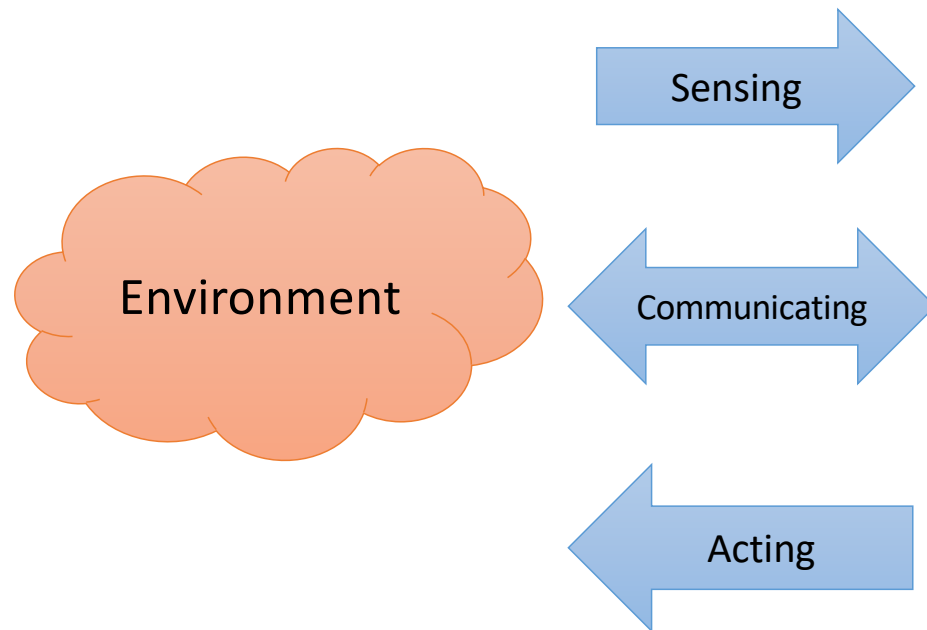
https://youtu.be/G-WdDeQ4TKw?si=EDGLXT0GE33VU_-w

Example: Space Exploration

Source: <https://pixabay.com/illustrations/martian-monster-green-1674016/>

Source: www.nasa.gov

AI: Building A Rational Agent



Perception
Learning
Modeling
Reasoning
Planning and Acting
Decision Making



Agent graphics source: ChatGPT4o, accessed 8 Aug 2024

A Planning Problem

- Assumption:

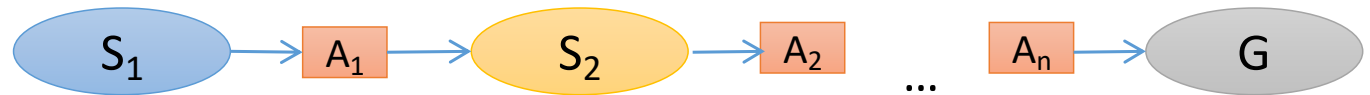
- Agent in task Environment

- Definition:

- States in an environment, with Initial State
- Actions available in a state
- Effects of applying an action
- Goal test to see if objective is met

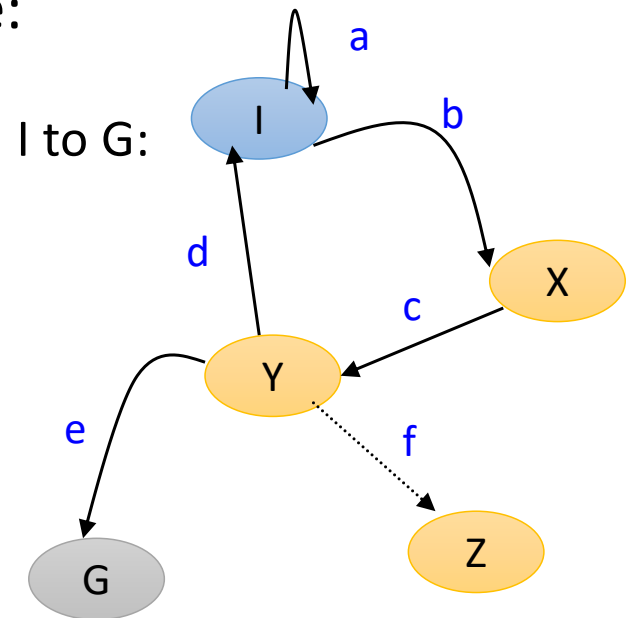
- Solution:

- Start from Initial State, find action sequence whose effects (resulting states) lead to the goal state



Example:

Plan from I to G:





Solving Planning Problems

- Planning Problem or Model
 - Appropriate abstraction of states, actions, effects, and goals (and costs and values)
- Planning Algorithm
 - Input: a problem
 - Output: a solution in the form of an **action sequence** (a **plan**)
- Planning Solution
 - A **plan** or **path** from the initial state(s) to the goal state(s)
 - Any path; OR
 - An **optimal** path wrt to costs or values
 - A **goal state** that satisfies certain properties

Planning Problem Types

Human Factors!

Problem Feature	Simple	Complex
States	Fully observable	Partially observable
Actions	Discrete	Continuous
Effects	Deterministic	Non-deterministic or Uncertain
Goals	Deterministic	Ordered or graded
Environment	Static	Dynamic
Agent	Single	Multiple

Real-Life Problems!

Costs, preferences, horizon, changes, etc.

Source: (GNT) Chapter 2, Ghallab, M., D. Nau, and P. Traverso, *Automated planning: Theory and practice*. 2004: Morgan Kaufmann



AI Planning and Sequential Decision Making

An intelligent agent needs to:

The **Actor's View of Planning**:

How to **plan to act** effectively in the real world?

How to **act to plan** effectively in the real world?

- make rational decisions
 - What does rational mean?
 - What are the decision objectives and guiding values?
- plan a sequence of actions to achieve some objective
 - How to learn to take actions optimally when there is uncertainty? change?
 - How to scale it up to large problems?
- act appropriately when there are other agents around
 - How to act when the other agents are also “thinking” and optimizing for themselves?
 - How to function and behave in a **responsible** manner in a human society?



Course Syllabus

- Classical planning (Review)
- Decision theory
- Probabilistic planning and decision making
- Reinforcement learning (Planning + Learning)
- Game theory and multi-agent decision making
- Large foundational models in planning
- And other new trends ..

The BIG Question

- With the recent advancements of Large Foundational Models:

Do we still need to learn
Sequential Decision Making techniques???

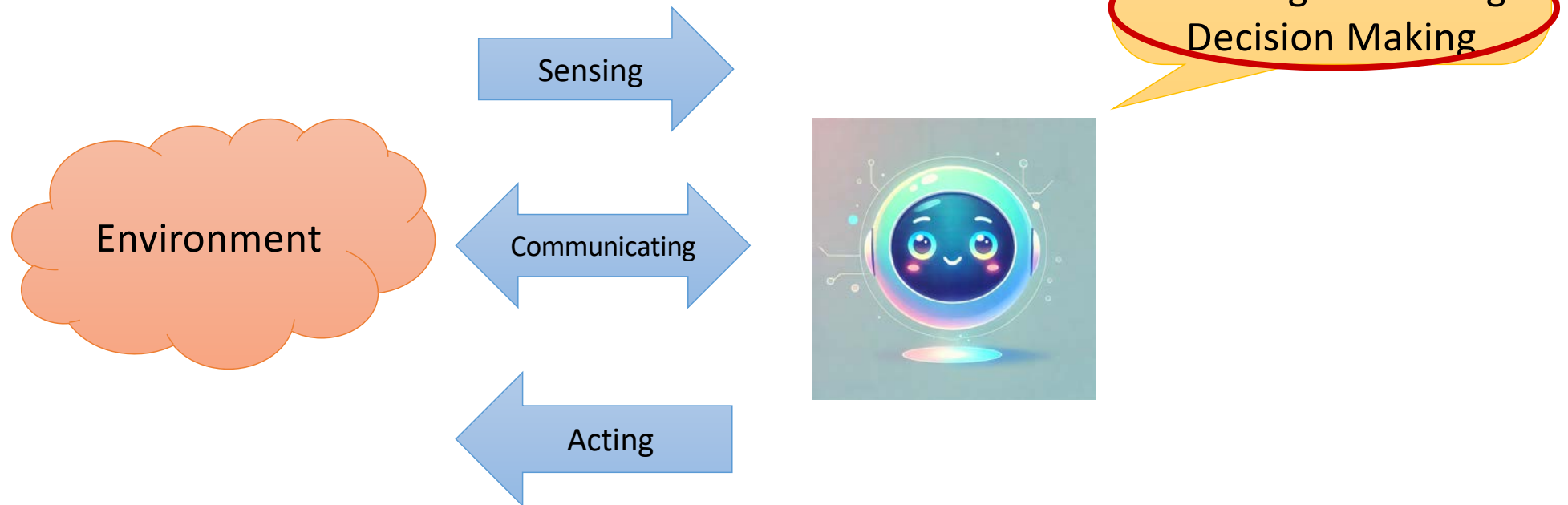
Agentic AI as sequential decision making ...



What is the State of the Art?

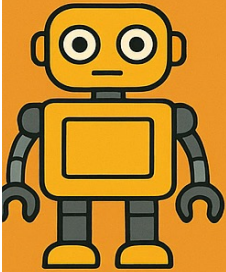
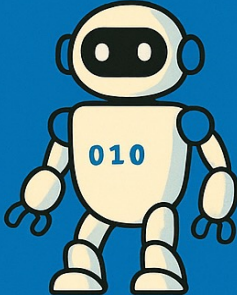
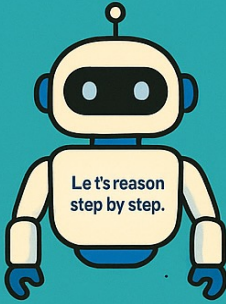
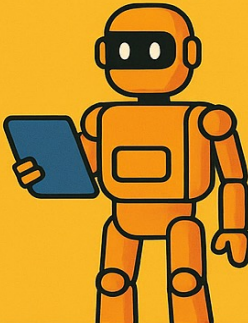
AI Planning and Decision Making: Now and Future

AI: Building A Rational Agent



Agent graphics source: ChatGPT4o, accessed 8 Aug 2024

A Brief History of Agents

1980s Symbolic AI agent	2000s Reinforcement Learning agent	2020s LLM agent	2030s Tool-using multi-agent
			
Block A is on block B. I want A on the table. must first clear A, then pick it up and move it."	[0.12, 1.5, -0.3] → action: turn left	"Let's reason step by step: If the user wants a restau- rant, I should find optio- then suggest reviews..."	To solve this: Search → extract → summarize → email result. Delegate steps to other agents.
Logic rules, symbolic planning	Learn by trial & error reward signals	Language-based reasoning, pretrained models	Planning + coordination, uses external tools or APIs



Symbolic Agent: Logical Inference

Symbolic AI and Logical Reasoning



Reinforcement Learning Agent: Trial and Error

Trial and Error for Long Term Optimal Policy

Source:

<https://youtu.be/gn4nRCC9TwQ?si=Ro3ZeHgHOj9k9akg>



Large Foundation Model Agent: ReACT

Reasoning and Acting via Language



Multi-Agent Collaboration

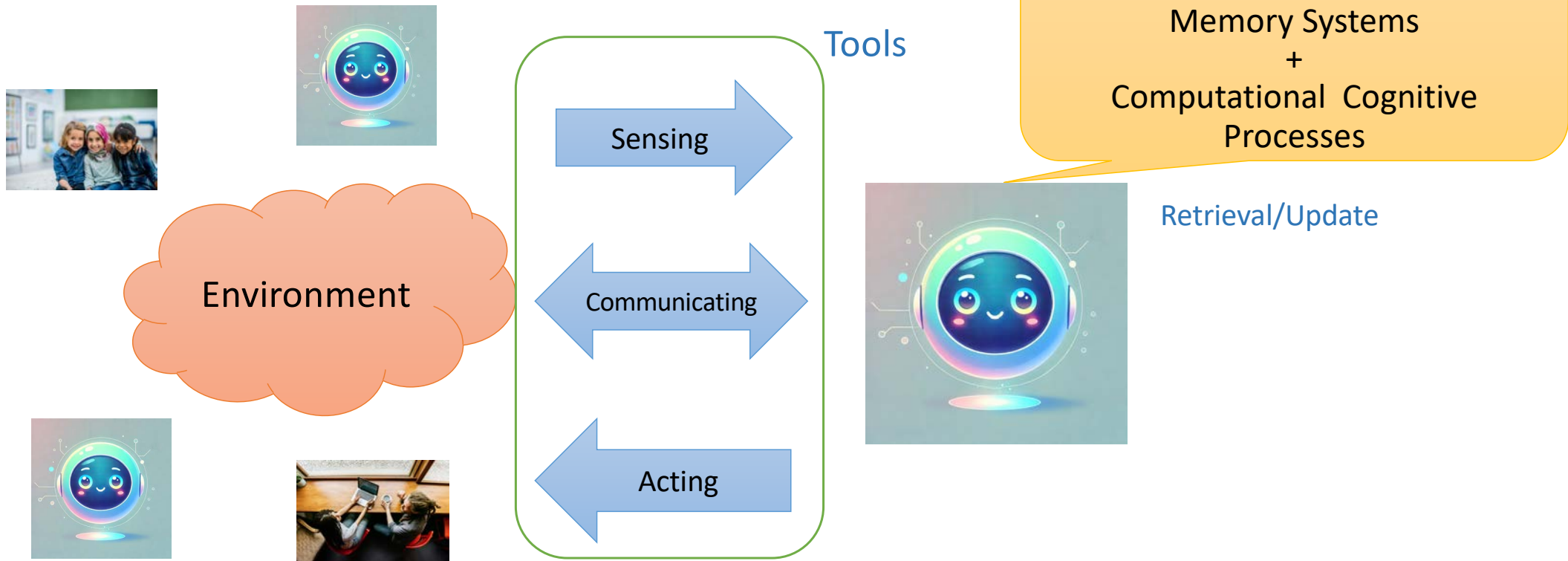
Multi-agent communication and coordination



Agentic AI in Practice Today

Andrew Ng Explores The Rise Of AI Agents And Agentic Reasoning | BUILD 2024 Keynote

Agentic AI Landscape



Overview

Agent graphics source: ChatGPT4o, accessed 8 Aug 2024

<https://www.forbes.com/councils/forbestechcouncil/2025/03/11/the-rise-of-the-one-person-unicorn-how-ai-agents-are-redefining-entrepreneurship/>

Overview <https://www.economist.com/business/2025/08/11/how-ai-could-create-the-first-one-person-unicorn>

<https://www.straitstimes.com/opinion/how-ai-could-create-the-first-one-person-unicorn>



Responsible AI Planning and Decision Making

- Toward **Human-Aware** AI Systems
 - AI working for, working with, and working alongside Humans
 - Human-AI collaboration
- Toward **Trustworthy** AI Systems
 - Natural interaction and effective collaboration
 - Fairness, accountability, and transparency
 - Robustness, resilience, privacy and security
 - Social, ethical, governance and regulatory considerations



Beyond Technical Challenges

- **Domain challenges**
 - Involving deep domain knowledge and operational issues
 - Interacting conditions, processes, and goals
- **User challenges**
 - Different skill levels and preferences
 - Varying usage patterns and cognitive biases
- **Economic challenges**
 - High implementation costs
 - Unclear market viability and scalability
- **System challenges**
 - Uncertain and changing information, processes, environments
 - Evolving IT and communication systems



Common Principles of Responsible AI

How to incorporate these considerations into “rational” decision making?

- Ensure safety
- Respect privacy
- Ensure fairness
- Promote trust
- Establish accountability
- Provide transparency
- Attribute responsibility
- Reflect diversity/inclusion
- Support equality
- Facilitate collaboration
- Uphold human rights and values
- Limit harmful uses of AI
- ...

Ref: RN: Chapter 27

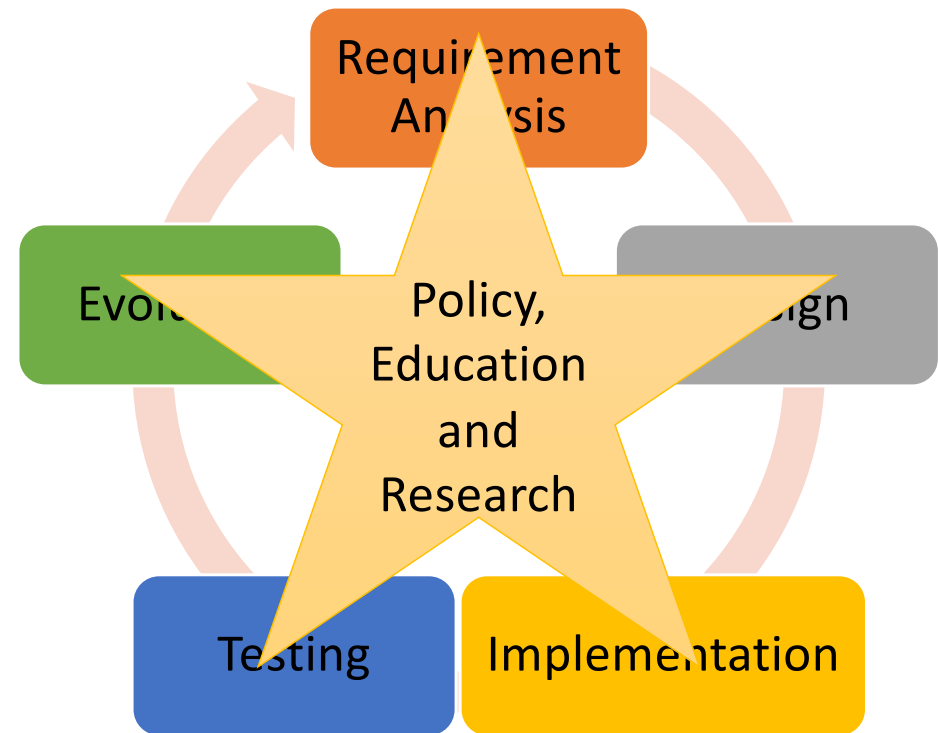


Making Responsible Decisions

- Main issue:
 - Value gained from **accurate results** balanced against additional efforts to incorporate **responsible feature(s)** – transparency, privacy, fairness, safety & robustness, human factors, governance & regulation
- Examples:
 - diagnosis recommendations, risk predictions, optimal policies, assistive guidance, recruitment interviews, social assistance schemes, university admissions, population censuses, consumer behaviors, scientific research
- If you are a developer, user, or owner/manager/regulator:
 1. What is <responsible feature>? Why is it important?
 2. What can be done to ensure trust (in AI decision systems)?
 3. What are the techniques and tools available?
 4. What is the tradeoff between accuracy and <responsible feature> of a trusted agent?
 5. What are the implications?
 6. Who should make the decisions? When?

When to Make Responsible Decisions?

- In AI design and application?
- In software engineering process?
- In deployment and maintenance?
- Who should make decisions?
- What are current practices?



AI Regulation on the Horizon

https://www.gov.cn/zhengce/202307/content_6892001.htm

<https://artificialintelligenceact.eu>

<https://www.whitehouse.gov/ostp/ai-bill-of-rights/>

<https://www.nature.com/articles/d41586-023-02491-y>

[https://www.hsa.gov.sg/docs/default-source/hprg-mdb/guidance-documents-for-medical-devices/gl-04-r3-regulatory-guidelines-for-software-medical-devices---a-life-cycle-approach-\(2024-mar\)-pub.pdf?sfvrsn=bbb0bdd2_4](https://www.hsa.gov.sg/docs/default-source/hprg-mdb/guidance-documents-for-medical-devices/gl-04-r3-regulatory-guidelines-for-software-medical-devices---a-life-cycle-approach-(2024-mar)-pub.pdf?sfvrsn=bbb0bdd2_4)

Example: AI in Education



<https://www.weforum.org/agenda/2024/02/with-generative-ai-we-can-reimagine-education-and-the-sky-is-the-limit/>

<https://www.unesco.org/en/articles/how-generative-ai-reshaping-education-asia-pacific>

Example: AI in Health Care

Home and Community

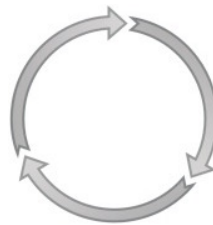
Policy, Regulation
and Governance

Clinic and Hospital

**Global Health,
Population Health,
Public Health**

Workflow and Process Optimization

Prevention and Care Support



**Scientific discovery
and experimentation**

Monitoring and Communication

Diagnosis and Treatment

Overview



Example: AI in Finance and Business

AI for Good!

nature communications

Explore Content ▾ Journal Information ▾ Publish With Us ▾

nature > nature communications > perspectives > article

Perspective | [Open Access](#) | Published: 13 January 2020

The role of artificial intelligence in achieving the Sustainable Development Goals

Ricardo Vinuesa  Hossein Azizpour Inês Leite Madeline Ralaam Virginia Dignum Sami Domisch  **unESCO** **ENGLISH**

Artificial intelligence In action Stories & Ideas Resources

Artificial Intelligence and the Futures of Learning

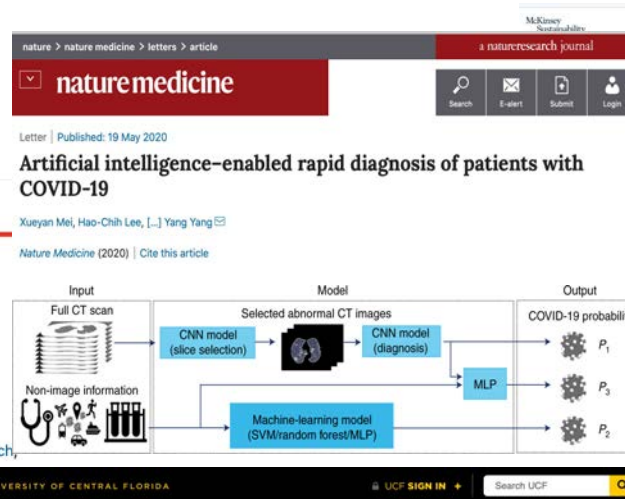


The Artificial Intelligence and the Futures of Learning project builds on the Recommendation on the Ethics of Artificial Intelligence to be adopted at the 41st session of the UNESCO General Conference and will follow up on the



Causality, Creativity and Imagination: New Frontiers in Planning

Sridhar Mahadevan (Adobe Research and University of Massachusetts Amherst)



UCF TODAY

SECTIONS

COLLEGES & CAMPUS NEWS

Where Artificial Intelligence Meets Urban Planning

First-year doctoral student is disseminating an AI program capable of changing the world of urban planning.

BY SIMONE ROUSSEAU | OCTOBER 6, 2020



Urban planning is a multi-faceted discipline in which planners collect data points from areas to determine if a store, hospital or other types of buildings are likely needed in that area.

MT Technology Review

Topics Magazine Newsletters Events

Artificial intelligence

AI planners in Minecraft could help machines design better cities

A competition to see which AI produces the best settlements in the game is helping to explore new techniques for urban planning.

by Will Douglas Heaven

September 22, 2020

< Back to Sustainability Blog

How AI can unlock a \$127B opportunity by reducing food waste

Scaling up AI's deep learning technology with a circular-economy vision could revolutionize the way food is grown, designed, purchased, and enjoyed.





By Clemmie Magnin

Managing Partner of McKinsey in France. Leads our work in the Consumer Packaged Goods and Retail Practices in Africa, Europe, and the Middle East, the agriculture sector in

 NATIONAL GEOGRAPHIC

March 27, 2019
[www.natgeo.com](#)

LOGIN C



Steam and smoke rise from the cooling towers and chimneys of a power plant. Artificial intelligence is being used to prove the case that plants that burn carbon-based fuels aren't profitable.

PHOTOGRAPH BY ROBB KENDRICK, NAT GEO IMAGE COLLECTION

NEWS

How artificial intelligence can tackle climate change

The biggest challenge on the planet might benefit from machine learning to help with solutions. Here are a just a few.

BY JACKIE SNOW



PUBLISHED JULY 18, 2019 • 6 MIN READ



Homework

- If you have not taken CS3263 before: (mandatory)
- Review Lecture (on Canvas):
 - Classical (Symbolic) Planning
 - As search and logical inference



Homework

1. Watch the Videos (Pages 36–40):

- Symbolic AI Agents
- Reinforcement Learning Agents
- Large Foundational AI Agents
- Multi-agent Collaboration
- Andrew Ng on Agentic AI in Practice

2. Research Further:

- What AI agents are commonly used in practice?
- What are the latest platforms or frameworks for building AI agents?
- What challenges or issues arise in using AI agents?

3. Deliverable:

- Prepare brief notes summarizing your findings for each question.
- Be ready to share and discuss in the Forum and/or the tutorial sessions